Know3D: Prompting 3D Generation with Knowledge from Vision-Language Models

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Research Questions

- How can vision-language models provide semantic guidance for controlling back-view generation in single-view 3D synthesis?
- What is the most effective way to transfer knowledge from multimodal diffusion models to 3D generation networks?
- Can language-controllable back-view generation improve the plausibility and user alignment of 3D assets?

Core Idea

Single-view 3D generation suffers from stochastic, uncontrollable back-view synthesis that often produces implausible geometries. Know3D addresses this by using a VLM-diffusion bridge: it fine-tunes Qwen-Image-Edit to generate semantically-guided back-view images from text prompts, then extracts intermediate hidden states from the diffusion transformer during denoising. These hidden states contain rich spatial-semantic information that gets injected into the 3D generation model as conditioning signals. This transforms random back-view hallucination into controllable, semantically-aware 3D synthesis.

Key Formula

Stochastic Prompt Construction

$$P = P_{view} \oplus P_{back}$$

P_view is fixed prompt for 180° camera rotation, P_back is randomly sampled from component descriptions with 0.5 probability. This enables learning both unconditional and semantically controlled back-view generation.

Conditional Flow Matching Loss

$$\mathcal{L}_{CFM}(\phi) = \mathbb{E}_{t,\, Z_{back},\, \epsilon} \| v_\phi(Z_t, t, H_{vlm}, Z_{front}) - (\epsilon - Z_{back}) \|_2^2$$

Z_t is noisy latent at timestep t, H_vlm are VLM hidden states, v_φ predicts velocity field. This trains the diffusion model to generate back-view conditioned on front view and text.

Key Figure

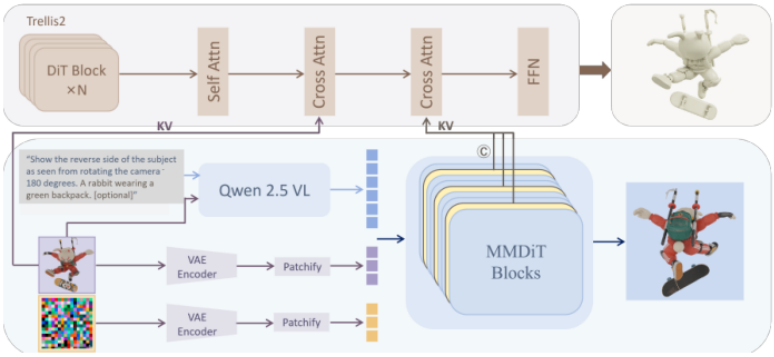

Fig. 3: Overview. Given an input image and a text prompt, Know3D leverages Qwen-Image-Edit-2511 [56], which integrates Qwen2.5-VL with a Diffusion Transformer (DiT). Qwen2.5-VL [2] encodes the multimodal inputs to guide a latent denoising process, enabling semantically aligned backside generation. The latent hidden states from intermediate DiT layers, which embed rich semantic and structural priors, are subsequently injected as conditioning signals into the 3D generation model to guide complete shape synthesis.

3.2 Prompting 3D Generation with VLMs

In this section, we introduce how to use image features as an intermediate medium to prompt knowledge from vision-language models (VLMs) for 3D generation models. The more straightforward approach is to directly inject the images generated by Qwen-Image. However, this process involves VAE decoding and the re-extraction of features by DINOv3 [41], making the workflow relatively cumbersome. Moreover, it relies on high-precision pixel-level restoration. If the quality of the generated images is insufficient, erroneous results will directly impact the 3D generation process. An ideal feature should possess the following characteristics: (1) sufficient spatial awareness to facilitate learning by 3D generation models, and (2) a certain degree of semantic awareness and robustness. We found that the hidden states of the intermediate layers of MMDiT inherently possess strong spatial awareness and rich semantic information [19, 26], enabling them to better guide 3D generation.

Knowledge Extraction and Prompting. Qwen2.5-VL [2] encodes the input front-view image and text prompt into high-level semantic features, while the VAE encoder extracts visual features from the front-view input. These representations guide the MMDiT through the full iterative denoising process. We extract intermediate latent hidden states from n MMDiT layers at a specific denoising timestep t [1, 19, 46], and concatenate these layer-wise features to form the structural-semantic conditioning signal $H_{\text{DiT}} = \text{Concat}(h^{(1)}, h^{(2)}, \ldots, h^{(n)})$.

Building upon TRELLIS2 [59], we design a parallel cross-attention branch for H_{DiT} injection. We retain the backbone's original self-attention and image-

Figure 3: Shows the complete Know3D architecture with VLM-diffusion bridge extracting hidden states from intermediate DiT layers for 3D generation conditioning.

Limitations

- Structural robustness still influenced by underlying multimodal foundation model capabilities
- When multimodal foundation model cannot understand instructions, 3D generation may produce incorrect shapes
- Performance depends on quality of fine-tuned Qwen-Image-Edit model for back-view generation
- Limited to back-view control rather than full 360-degree semantic control